Project Title: Fuzzy Logic-Based Smart Irrigation Control System

Description:
This project uses fuzzy logic to decide the amount of water required for plants based on soil moisture, temperature, and humidity. MATLAB Fuzzy Logic Toolbox is used to design the membership functions, rules, and controller output.

Submission: Week 12 (31^{st} July)

Submission Documents: Report that contains code snippets and illustration of every step/process.

Group Members:

No	Name	Student ID
1	Kheshwenda A/L Narasimban	24003160
2	Anuar Afiq Bin Arfahairy	24002263
3	Fawwaz 'Arash Bin Mohamad Fakhri	24001456

Define System Input and Output:
Input: SoilMoisture, Temperature, Humidity
Output: WaterAmount

Define range of variable input:
Input 1 (SoilMoisture): 0% to 100%
Input 2 (Temperature): 0°C to 50°C
Input 3 (Humidity): 0% to 100%

Define Variable system structure:

Variable Type	Variable Name	Range	Status words / Action Words
Input 1	SoilMoisture	0% to 100%	Dry, Moist, Wet
Input 2	Temperature	0°C to 50°C	Cold, Normal, Hot
Input 3	Humidity	0% to 100%	Low, Medium, High
Output	WaterAmount	0% to 100%	Zero, Low, Medium, Large

Define Rule-Base:

1. If **SoilMoisture** is Dry and **Temperature** is Cold and **Humidity** is Low, then **WaterAmount** is Medium.
2. If **SoilMoisture** is Moist and **Temperature** is Cold and **Humidity** is Low, then **WaterAmount** is Low.
3. If **SoilMoisture** is Wet and **Temperature** is Cold and **Humidity** is Low, then **WaterAmount** is Zero.
4. If **SoilMoisture** is Dry and **Temperature** is Normal and **Humidity** is Low, then **WaterAmount** is Large.
5. If **SoilMoisture** is Moist and **Temperature** is Normal and **Humidity** is Low, then **WaterAmount** is Medium.
6. If **SoilMoisture** is Wet and **Temperature** is Normal and **Humidity** is Low, then **WaterAmount** is Zero.
7. If **SoilMoisture** is Dry and **Temperature** is Hot and **Humidity** is Low, then **WaterAmount** is Large.
8. If **SoilMoisture** is Moist and **Temperature** is Hot and **Humidity** is Low, then **WaterAmount** is Large.
9. If **SoilMoisture** is Wet and **Temperature** is Hot and **Humidity** is Low, then **WaterAmount** is Zero.
10. If **SoilMoisture** is Dry and **Temperature** is Cold and **Humidity** is Medium, then **WaterAmount** is Medium.
11. If **SoilMoisture** is Moist and **Temperature** is Cold and **Humidity** is Medium, then **WaterAmount** is Low.
12. If **SoilMoisture** is Wet and **Temperature** is Cold and **Humidity** is Medium, then **WaterAmount** is Zero.
13. If **SoilMoisture** is Dry and **Temperature** is Normal and **Humidity** is Medium, then **WaterAmount** is Medium.
14. If **SoilMoisture** is Moist and **Temperature** is Normal and **Humidity** is Medium, then **WaterAmount** is Low.
15. If **SoilMoisture** is Wet and **Temperature** is Normal and **Humidity** is Medium, then **WaterAmount** is Zero.

16. If **SoilMoisture** is Dry and **Temperature** is Hot and **Humidity** is Medium, then **WaterAmount** is Large.
17. If **SoilMoisture** is Moist and **Temperature** is Hot and **Humidity** is Medium, then **WaterAmount** is Medium.
18. If **SoilMoisture** is Wet and **Temperature** is Hot and **Humidity** is Medium, then **WaterAmount** is Zero.
19. If **SoilMoisture** is Dry and **Temperature** is Cold and **Humidity** is High, then **WaterAmount** is Low.
20. If **SoilMoisture** is Moist and **Temperature** is Cold and **Humidity** is High, then **WaterAmount** is Zero.
21. If **SoilMoisture** is Wet and **Temperature** is Cold and **Humidity** is High, then **WaterAmount** is Zero.
22. If **SoilMoisture** is Dry and **Temperature** is Normal and **Humidity** is High, then **WaterAmount** is Medium.
23. If **SoilMoisture** is Moist and **Temperature** is Normal and **Humidity** is High, then **WaterAmount** is Zero.
24. If **SoilMoisture** is Wet and **Temperature** is Normal and **Humidity** is High, then **WaterAmount** is Zero.
25. If **SoilMoisture** is Dry and **Temperature** is Hot and **Humidity** is High, then **WaterAmount** is Medium.
26. If **SoilMoisture** is Moist and **Temperature** is Hot and **Humidity** is High, then **WaterAmount** is Low.
27. If **SoilMoisture** is Wet and **Temperature** is Hot and **Humidity** is High, then **WaterAmount** is Zero.

Define Member Functions:

1. Input 1 (SoilMosture)
    - Dry: [0 0 25 45] - Trapezoid
    - Moist: [30 50 70] - Triangular
    - Wet: [55 75 100 100] - Trapezoid
2. Input 2 (Temperature)
    - Cold: [0 0 12 22] - Trapezoid
    - Normal: [15 25 35] - Triangular
    - Hot: [28 38 50 50] - Trapezoid
3. Input 3 (Humidity)
    - Low: [0 0 25 45] - Trapezoid
    - Medium: [30 50 70] - Triangular
    - High: [55 75 100 100] - Trapezoid
4. Output (WaterAmount)
    - Zero: [0 0 10 25] - Trapezoid
    - Low: [15 35 55] - Triangular
    - Medium: [45 65 85] - Triangular
    - Large: [75 90 100 100] - Trapezoid

Fuzzy Inference System Architecture:
We chose the Mamdani method because its rules are easy to understand, using everyday language. This makes it simple to set up the system rules based on how humans would naturally think about the weather and plant care.

When the system checks multiple conditions at once, for example: soil is dry AND temperature is hot, it uses the Minimum or Product method to combine those values and calculate the final result.

Defuzzication:
To convert fuzzy output into *WaterAmount,* **Centroid** defuzzication method is used.

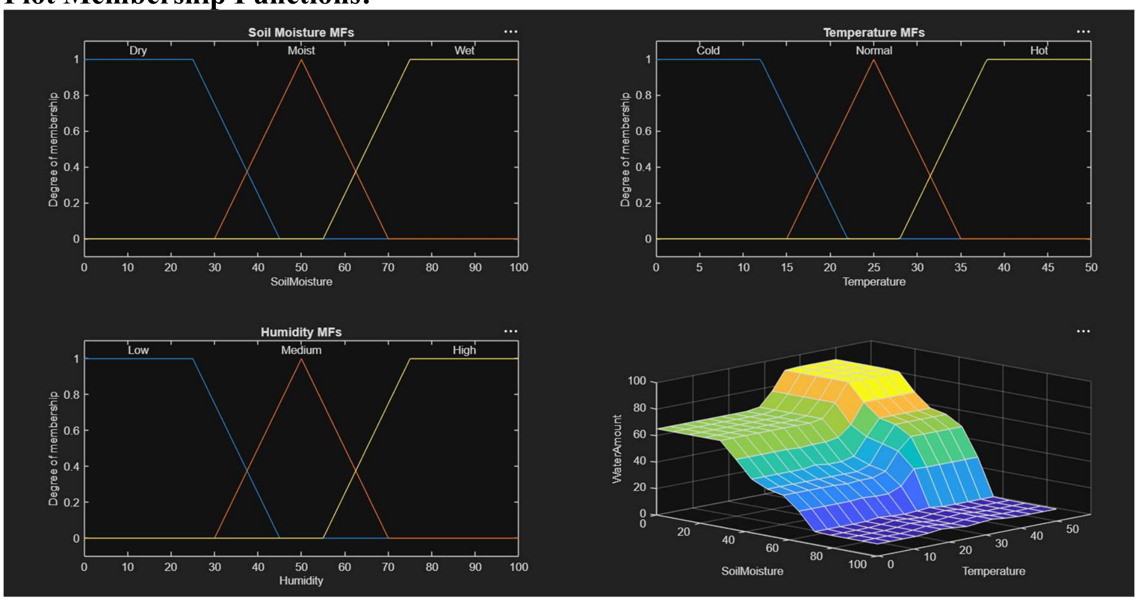
Design Script:

```
clear; clc; close all;

% Construct FIS
SmartIrrigationControlSystem = mamfis(Name="SmartIrrigationControlSystem");

% Input 1: Soil Moisture (0% to 100%)
SmartIrrigationControlSystem = addInput(SmartIrrigationControlSystem, [0 100], Name="SoilMoisture");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "SoilMoisture", "trapmf", [0 0 25 45], Name="Dry", VariableType="input");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "SoilMoisture", "trimf", [30 50 70], Name="Moist", VariableType="input");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "SoilMoisture", "trapmf", [55 75 100 100], Name="Wet", VariableType="input");

% Input 2: Temperature (0°C to 50°C)
SmartIrrigationControlSystem = addInput(SmartIrrigationControlSystem, [0 50], Name="Temperature");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "Temperature", "trapmf", [0 0 12 22], Name="Cold", VariableType="input");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "Temperature", "trimf", [15 25 35], Name="Normal", VariableType="input");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "Temperature", "trapmf", [28 38 50 50], Name="Hot", VariableType="input");

% Input 3: Humidity (0% to 100%)
SmartIrrigationControlSystem = addInput(SmartIrrigationControlSystem, [0 100], Name="Humidity");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "Humidity", "trapmf", [0 0 25 45], Name="Low", VariableType="input");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "Humidity", "trimf", [30 50 70], Name="Medium", VariableType="input");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "Humidity", "trapmf", [55 75 100 100], Name="High", VariableType="input");
```

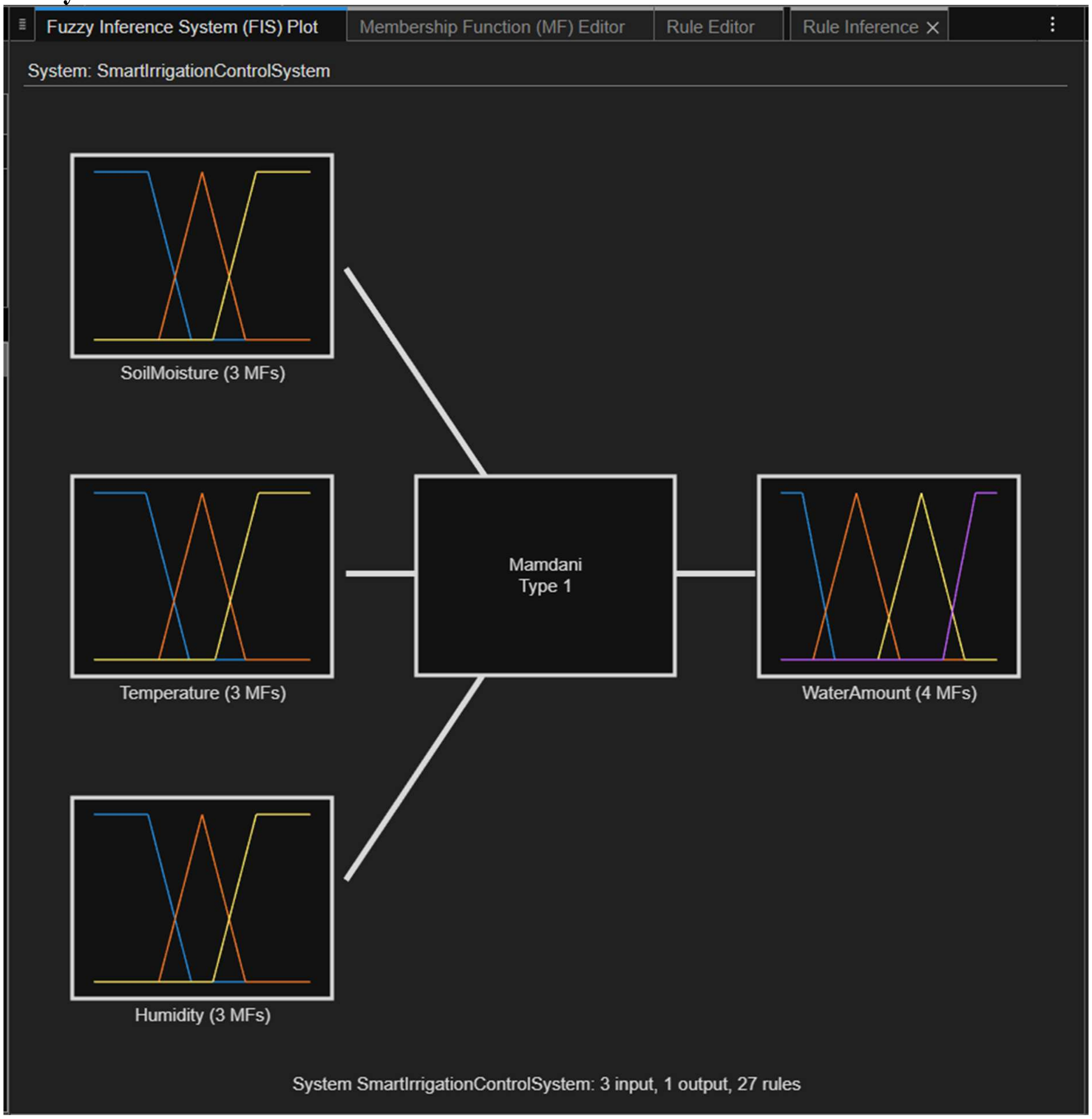
```
% Output 1: Water Amount (0% to 100%)
SmartIrrigationControlSystem = addOutput(SmartIrrigationControlSystem, [0 100],
Name="WaterAmount");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "WaterAmount", "trapmf", [0 0
10 25], Name="Zero", VariableType="output");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "WaterAmount", "trimf", [15 35
55], Name="Low", VariableType="output");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "WaterAmount", "trimf", [45 65
85], Name="Medium", VariableType="output");
SmartIrrigationControlSystem = addMF(SmartIrrigationControlSystem, "WaterAmount", "trapmf", [75
90 100 100], Name="Large", VariableType="output");

% Rules
rulesList = [
    1 1 1 3 1 1;
    2 1 1 2 1 1;
    3 1 1 1 1 1;
    1 2 1 4 1 1;
    2 2 1 3 1 1;
    3 2 1 1 1 1;
    1 3 1 4 1 1;
    2 3 1 4 1 1;
    3 3 1 1 1 1;
    1 1 2 3 1 1;
    2 1 2 2 1 1;
    3 1 2 1 1 1;
    1 2 2 3 1 1;
    2 2 2 2 1 1;
    3 2 2 1 1 1;
    1 3 2 4 1 1;
    2 3 2 3 1 1;
    3 3 2 1 1 1;
    1 1 3 2 1 1;
    2 1 3 1 1 1;
    3 1 3 1 1 1;
    1 2 3 3 1 1;
    2 2 3 1 1 1;
    3 2 3 1 1 1;
    1 3 3 3 1 1;
    2 3 3 2 1 1;
    3 3 3 1 1 1
];

SmartIrrigationControlSystem = addRule(SmartIrrigationControlSystem, rulesList);

% Plot Membership Functions
figure('Name', 'Membership Function Plots');
subplot(2,2,1); plotmf(SmartIrrigationControlSystem, 'input', 1); title('Soil Moisture MFs');
subplot(2,2,2); plotmf(SmartIrrigationControlSystem, 'input', 2); title('Temperature MFs');
subplot(2,2,3); plotmf(SmartIrrigationControlSystem, 'input', 3); title('Humidity MFs');
subplot(2,2,4); plotmf(SmartIrrigationControlSystem, 'output', 1); title('Water Amount MFs');
```

```
% Test Case Simulation: Dry Soil (15%), Hot Temp (38°C), Low Humidity (20%)
testInput = [15, 38, 20];
calculatedWater = evalfis(SmartIrrigationControlSystem, testInput);
fprintf('Simulation Test\n');
fprintf(' \n');
fprintf('Input Conditions  : Soil Moisture = %d%%, Temp = %d°C, Humidity = %d%%\n', testInput(1),
testInput(2), testInput(3));
fprintf('Controller Output : Water Amount = %.2f%%\n', calculatedWater);

gensurf(SmartIrrigationControlSystem); % Generates 3D Surface View
ruleview(SmartIrrigationControlSystem); % Generates Rule Viewer
```

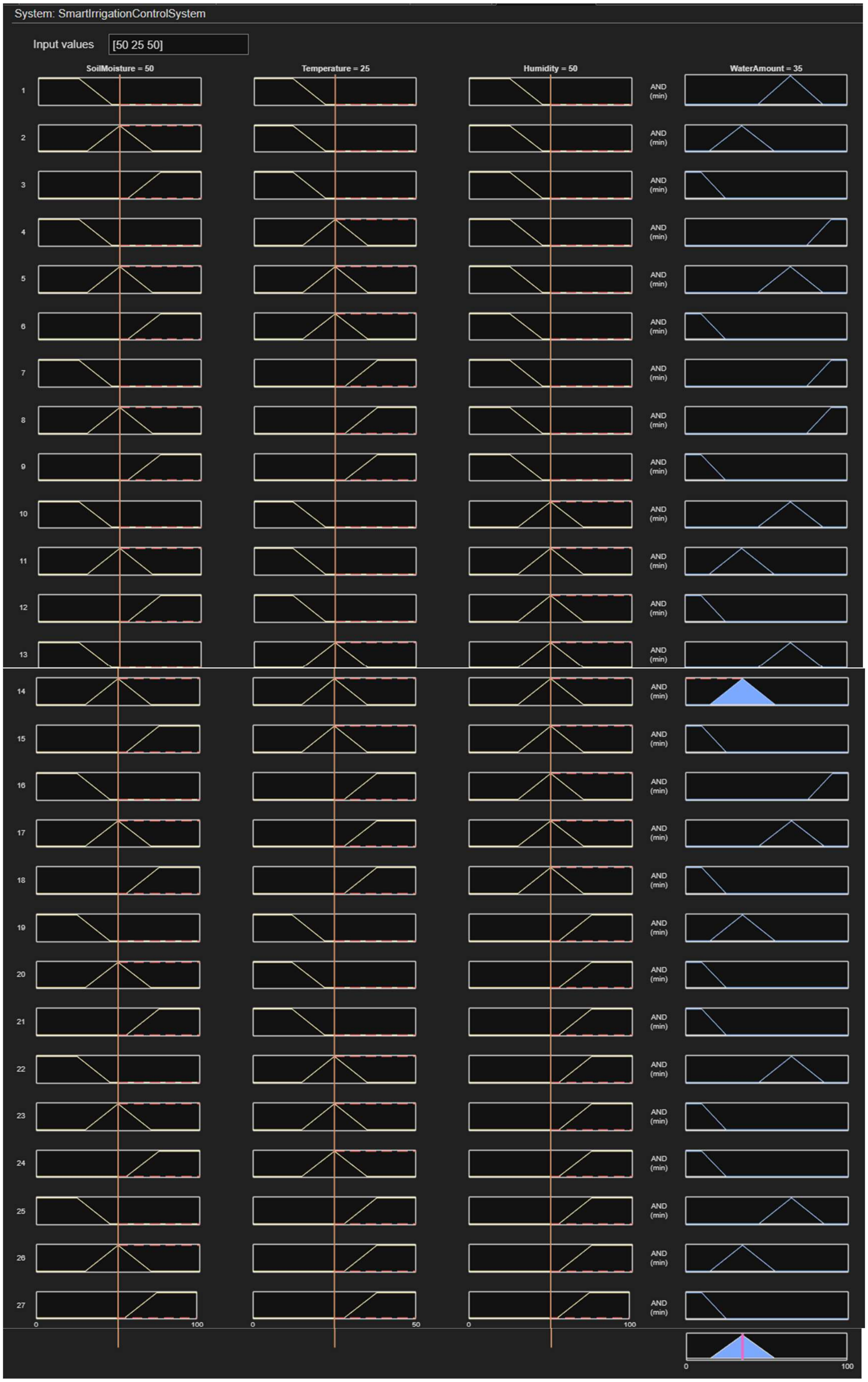
Plot Membership Functions:



Fuzzy Inference Plot:



Rule Viewer:


System: SmartIrrigationControlSystem
Input values
[50 25 50]
SoilMoisture = 50
Temperature = 25
Humidity = 50
WaterAmount = 35
AND (min)

Simulation Test:
Soil Moisture: 15%
Temperature: 38%
Humidity: 20%

Output:

```
Command Window
Simulation Test

Input Conditions  : Soil Moisture = 15%, Temp = 38°C, Humidity = 20%
Controller Output : Water Amount = 90.98%
```